

Experimental quantification of coupled thermal-mechanical nonlinearities in a bolted joint

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Abstract

Coupled thermal-mechanical nonlinearities are an important consideration in structural engineering but less widely investigated effects within mechanical engineering; thermal and mechanical loads are typically considered decoupled, an approach that is insufficient when nonlinear behaviour arises. To determine the coupled thermal-mechanical effects on the dynamic behaviour of jointed systems, two systems are exposed to varying thermal loading conditions across an excitation frequency range around resonance, with comparative analysis attributing behaviour directly to a bolted joint interface. Analysing backbone curves, stiffness deviations, and cubic modal stiffness coefficients reveals that the jointed system undergoes a temperature-driven reversal in its nonlinear behaviour. This shift results from the competition between temperature-modulated geometric softening and contact micro-slip hardening at the joint interface. This provides evidence that the decoupled methodology may be insufficient for a bolted joint.

1 Introduction

Bolted joints are ubiquitous in structural assemblies because they maintain assembly flexibility whilst being able to carry large mechanical loads across environmental extremes [1]. This is particularly critical in space and satellite systems, where structures are routinely subjected to combined thermal and mechanical loads [2]. For example, a typical CubeSat in low Earth orbit undergoes rapid temperature cycles, ranging from roughly -40°C in eclipse to $+85^{\circ}\text{C}$ in sunlight [3], while simultaneously experiencing vibrations from internal components such as reaction wheels. In these environments, structural predictability is crucial, as even minor shifts in a system's dynamic properties can cause unexpected resonance with launch or operational loads, leading to localised mechanical fatigue or catastrophic structural failure.

Current research extensively explores the effects of thermal and mechanical loads, but usually does so in isolation or sequentially by distinguishing between environmental thermal effects and structural modal parameters [4, 5]. This decoupled approach is often sufficient for linear elastic systems, where temperature-dependent changes to boundary conditions, material properties, and contact conditions can be treated independently at each thermal state. In such cases, the mechanical vibrations do not alter the temperature distribution, and the thermal state does not introduce amplitude-dependent changes to the system's energy dissipation mechanisms. Note that the term *coupled* is used throughout this work to describe the concurrent application of thermal and mechanical loading, not two-way thermal-mechanical coupling in which vibration itself modifies the thermal field. However, for systems where nonlinearity is prevalent, this approach provides an incomplete picture, as superposition is invalidated by the nonlinearities. In these cases, nonlinearities may arise from the coupled effects of concurrent thermal-mechanical loads, which are difficult to predict using traditional decoupled methods [6]. While there are examples of direct coupling and reduced-order modelling [7], such approaches are often not tailored to joint interfaces.

Within structures that experience these combined loads, nonlinearities can originate from the contact interfaces at bolted joints, manifesting as variations in stiffness, energy dissipation, and complex stick-slip phenomena [8], influenced by contact interface mechanics and material-dependent thermal effects. While

extensive research has characterised the mechanical response of joints under constant-temperature conditions [9, 10, 11], there remains a lack of systematic investigation into combined loading effects. The closest prior work, by Molaie et al. [12], establishes a foundation for this area by investigating how varying thermal environments alter the nonlinear dynamics, specifically through stiffness, of a continuous bimetallic beam. While their work successfully captures macro-structural thermal dependencies, practical structural assemblies are heavily governed by localised contact interface interactions. These contact interface interactions lead to critical structural integrity issues in practice; specifically, joint-induced nonlinearities can result in accelerated fatigue and the progressive loss of bolt preload [13].

The present study experimentally investigates a simplified bolted lap-joint under controlled temperature cycling and vibrational loading. The primary objective is to quantify how temperature influences stiffness and the manifestation of nonlinearity. By comparing the jointed specimen to a monolithic baseline of identical geometry and material, the study isolates the behaviour driven specifically by the joint interface. This setup enables the characterisation of how thermal changes drive stiffness degradation and modal shifts, ultimately informing more reliable design criteria for engineering assemblies subject to coupled loads.

Based on prior studies of bolted joint dynamics and nonlinear vibration behaviour, it is anticipated that the jointed specimen will exhibit amplitude- and temperature-dependent nonlinear characteristics, in particular the shift in onset of nonlinear response through changes in preload, material properties, and contact interface conditions [14], that are absent or significantly reduced in the equivalent monolithic specimen [15]. By directly comparing jointed and equivalent monolithic configurations under identical thermal and mechanical excitation, this study aims to isolate joint-induced nonlinearities and provide experimentally grounded insight into how coupled thermal-mechanical loading affects dynamic response. The results are intended to inform improved modelling approaches and parameter identification routines for bolted joints operating in thermally variable environments, such as satellite and space structures.

2 Methodology

2.1 Specimens and Configuration

To isolate joint mechanics, two beams are used; one which includes a jointed interface, consisting of two beams with an overlapping bolted lap joint, and an equivalent monolithic beam that possesses the same single-span geometry while removing the joint. The beams are clamped at their ends to simulate fixed boundary conditions and are attached to the shaker plate at the base of the environmental chamber, allowing for excitation whilst maintaining stable thermal loading within the chamber's confines. The clamps were tightened to a torque of 3Nm using a calibrated torque wrench and were not adjusted during testing; the same clamp arrangement was used for both specimens. The bolted lap joint was assembled using an M5 bolt tightened to 1.5Nm. Accelerometers are mounted to the specimens equally spaced along their length so that averaging can be utilised to capture overall system response, and one is mounted to the base plate itself to capture base acceleration. The specimens for these experiments are pictured in Figures 1a and 1b.

This setup was influenced by the work of Brake and Reuß [16]; however, its design differs in three key ways. Firstly, the geometry has been designed with a thinner cross-section in mind, enhancing the sensitivity to joint nonlinearities, as well as its generally smaller footprint, so as to fit within the confines of the environmental chamber. Secondly, steel was chosen as a low-cost and readily available material, as opposed to the aluminium used in the Brake-Reuß beam. Thirdly, instead of being suspended with free ends, this setup utilises clamped-clamped conditions, as Bhattu et al. suggest will show more distinct modal behaviour [17].

2.2 Testing Methodology

First, impact hammer testing was performed to identify the dominant resonance frequencies of both the jointed and equivalent monolithic specimens, and to inform the selection of the excitation frequency range used in subsequent experiments. An instrumented impact hammer was used to apply impulsive excitation at

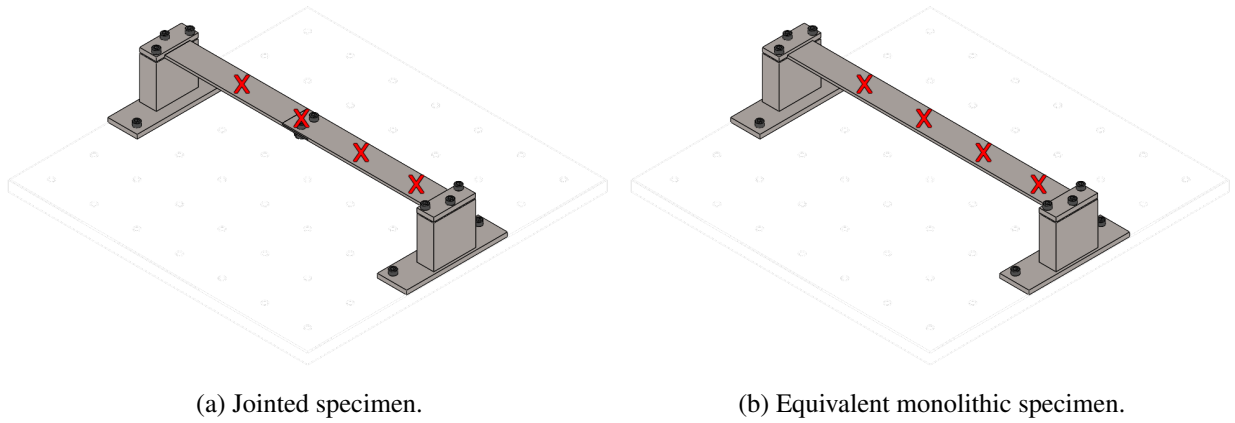


Figure 1: 3D representations of the specimens tested. Accelerometer positions are denoted by red X's along the beams.

a consistent location, and multiple impacts were applied for averaging. This enabled the identification of the frequency ranges associated with the fundamental mode of each beam.

Following the identification of the resonance frequencies for both specimens, the thermal-mechanical stepped-sine test campaign was conducted; the excitation frequency was incremented in discrete steps across a narrow frequency band spanning resonance, and at each frequency step, the specimen was excited using a single-frequency harmonic input at a fixed amplitude. The excitation was held constant for a prescribed dwell time, allowing the response to reach a steady-state condition. Once steady state was achieved, acceleration response data were captured and stored for subsequent analysis. The excitation frequency was then incremented to the next step, and the process was repeated until the full frequency range had been covered. By stepping the frequency rather than continuously sweeping, the response at each frequency could be treated as quasi-steady-state, improving the reliability of amplitude and phase measurements near resonance.

Testing was conducted at 16 discrete temperatures from -30°C to $+120^{\circ}\text{C}$ in 10°C increments, and across a frequency range of 5 Hz around each respective fundamental mode in 0.1 Hz steps. The full frequency-amplitude stepped-sine procedure was repeated in its entirety at each set point following thermal stabilisation, defined as the chamber thermocouple reading remaining within $\pm 0.5^{\circ}\text{C}$ of the target set point for a minimum soak period of 10 minutes. Once the full frequency range had been covered for a given excitation amplitude, the excitation amplitude is increased to its next predefined level, and the previous stepped frequency sequence is then repeated in its entirety at this new amplitude. This process continued until the complete set of excitation amplitudes had been tested. Once the full frequency-amplitude test set at a given temperature had been completed, the temperature was adjusted to the next target set point and the procedure was repeated. The entire experimental workflow is shown in Figure 2.

One concern regarding the accuracy of the readings was the lack of temperature-insensitive accelerometers, as one of the main challenges in thermal testing is the temperature dependency of accelerometer gain. To compensate for this, the measured signals were corrected using a third-order polynomial derived from the manufacturer's calibration data obtained at initial purchase, as shown in Figure 3. This correction was incorporated into the data processing workflow so that the temperature-dependent sensitivity deviations could be compensated for.

The stepped-sine shaker data were post-processed using a custom MATLAB workflow to extract time- and frequency-domain response metrics for each excitation amplitude, frequency step, and temperature. Each data file is segmented into discrete blocks corresponding to individual stepped-sine dwell periods, allowing analysis of the steady-state behaviour at each test condition.

These processed data served as the primary input for characterising the joint's nonlinear behaviour across varying thermal loads. Specifically, steady-state acceleration peaks were extracted to determine response transmissibility, which served as the fundamental measure of energy transfer and stiffness degradation. By focusing on steady-state transmissibility and stiffness deviation rather than the raw time series, the workflow

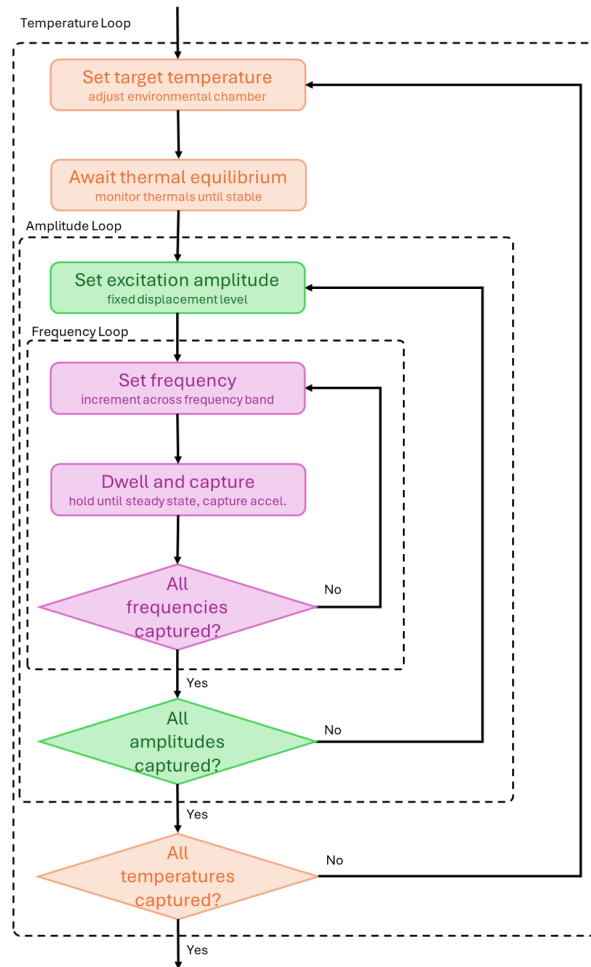


Figure 2: Experimental workflow showing the nested loops for temperature, amplitude, and stepped-sine frequency sweeps.

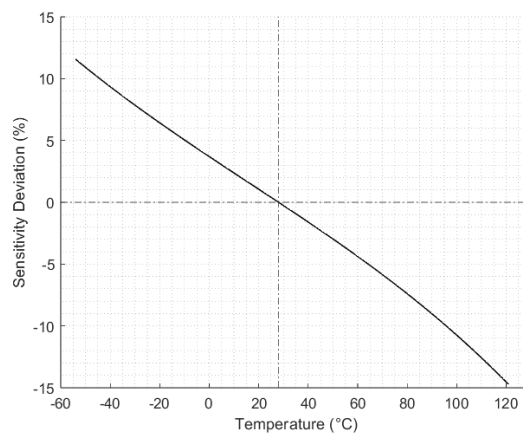


Figure 3: Sensitivity deviation with temperature of the 352C22 accelerometers, given by the manufacturer.

helped isolate thermally dependent nonlinear effects associated with the joint interface, facilitating reliable comparison across test conditions and joint configurations.

3 Results and Discussion

3.1 Mode Identification via Impact Hammer Testing

Impact hammer testing was performed at ambient temperature (20°C) prior to thermal cycling. Throughout this work, subscripts j and m are used to denote quantities associated with the jointed and monolithic specimens, respectively. The fundamental bending mode (Mode 1) was identified at $f_J = 116.94$ Hz for the jointed specimen and $f_M = 150.37$ Hz for the monolithic specimen, as shown in Figure 4. These tests were conducted at the lowest practicable excitation level to ensure the response remained within the linear regime, consistent with the low-amplitude limit of the stepped-sine data. Both frequency response functions exhibit additional higher-order bending modes above 200 Hz; however, the present study focuses on Mode 1, which produces the largest amplitude response under the stepped-sine excitation conditions and is most sensitive to joint-induced nonlinearity. Using the physical masses ($m_J = 0.233$ kg, $m_M = 0.228$ kg), the corresponding equivalent stiffnesses are $k_J = 125.8$ kN/m and $k_M = 203.7$ kN/m.

The bolted lap joint reduces equivalent stiffness by 38.2% relative to the monolithic beam. This is consistent with the compliance introduced by the lap joint geometry: the load path through the bolt shanks and contact patch replaces a continuous beam cross-section in the overlap region, reducing the effective second moment of area. The frequency ratio between specimens is consistent with this stiffness reduction, as expected from $f \propto \sqrt{k/m}$:

$$\frac{k_J}{k_M} = \frac{m_J}{m_M} \left(\frac{f_J}{f_M} \right)^2 = \frac{0.233}{0.228} \left(\frac{116.94}{150.37} \right)^2 = 0.618 \quad (1)$$

which confirms internal consistency between the measured frequencies and estimated equivalent stiffnesses, and demonstrates that the stiffness difference between specimens is captured by the shift in natural frequency alone. Equivalent damping ratios extracted from the half-power bandwidth are $\zeta_J = 0.00136$ and $\zeta_M = 0.00109$, a difference of 24.8% relative to the monolithic value. The jointed beam exhibits higher damping, consistent with frictional energy dissipation at the contact interface, a well-documented characteristic of bolted joints [8]. However, it is noted that the clamped boundary conditions may also contribute to energy dissipation, and the difference cannot be attributed solely to the lap joint. Furthermore, since both values are at the level of material damping, this difference produces no visually discernible separation between the two FRFs in Figure 4, and the distinction is only apparent through quantitative extraction.

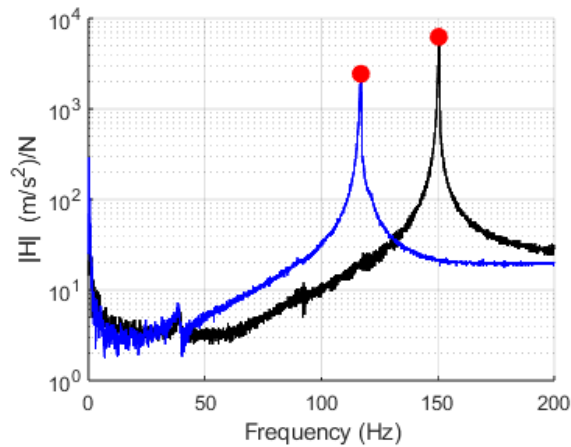


Figure 4: Frequency response functions from impact hammer testing showing Mode 1 resonance of the jointed (blue) and monolithic (black) specimens.

3.2 Cubic Modal Stiffness Quantification

To quantify the amplitude-dependent stiffness behaviour across all temperatures, the backbone curves are analysed within the Duffing oscillator framework [18]. For a Duffing oscillator with cubic stiffness nonlinearity, the squared resonance frequency varies linearly with the square of the displacement amplitude:

$$f_{\text{res}}^2(A, T) = f_0^2(T) + \alpha(T) A^2 \quad (2)$$

where $f_0(T)$ is the linear natural frequency at temperature T , A is the displacement amplitude (μm), and $\alpha(T)$ is the cubic modal stiffness coefficient. The sign of $\alpha(T)$ defines the net nonlinear character: $\alpha < 0$ denotes a softening response, whereas $\alpha > 0$ indicates a hardening response. The coefficient α carries units of $\text{Hz}^2/\mu\text{m}^2$, consistent with identification in the frequency-amplitude (modal) domain. This differs from the physical Duffing stiffness parameter, which carries units of N/m^3 ; conversion between the two requires knowledge of the modal mass. Accordingly, α is referred to here as the *cubic modal stiffness coefficient* rather than the physical Duffing stiffness. Equation (2) is fitted by least-squares to the measured (A, f_{res}) pairs at each temperature; the coefficient of determination R^2 indicates whether a single cubic term adequately describes the backbone.

3.3 Amplitude Dependence: Jointed versus Monolithic

To isolate the effect of the bolted interface at a fixed thermal condition, Figure 5 shows the backbone curves of both specimens at 20°C across the full range of tested acceleration amplitudes (0.022 - 0.384 g). The monolithic backbone tilts consistently leftward, indicating a softening nonlinearity in which the resonance frequency decreases with increasing amplitude. This behaviour is well-established for clamped-clamped beams and arises from von Kármán membrane-bending coupling: transverse deflection induces midplane stretching, generating a geometric restoring force that grows sub-cubically with displacement and reduces the effective stiffness at large amplitudes [19].

The jointed backbone at the same temperature tilts rightward, indicating a hardening response in which resonance frequency increases with amplitude. This reversal in nonlinear character is attributable entirely to the bolted interface: the monolithic baseline confirms that the beam geometry alone produces softening, so any hardening observed in the jointed system must originate at the contact. This amplitude-dependent hardening from frictional interfaces has been reported in prior studies of jointed structures under ambient conditions [9, 10, 11], and the present result confirms that it persists under coupled thermal loading.

Applying the framework detailed in Section 3.2, least-squares fitting yields cubic modal stiffness coefficients at 20°C of $\alpha_J = +1.71 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ (hardening) for the jointed beam and $\alpha_M = -5.00 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ (softening) for the monolithic beam. The opposing signs are a quantitative confirmation that the bolted interface reverses the nonlinear character relative to the monolithic baseline.

The normalised stiffness deviation $\Delta K/K_0$, where $\Delta K = K(A) - K_0$ and K_0 is the low-amplitude linear stiffness ($K_{0,J} = 125.8 \text{ kN/m}$, $K_{0,M} = 203.7 \text{ kN/m}$), characterises this behaviour over the tested displacement range. At 20°C the jointed beam ($K_{0,J}$) exhibits a characteristic dip-then-rise profile: $\Delta K/K_0$ begins at -0.908% ($\Delta K \approx -1,142 \text{ N/m}$) at $43.7 \mu\text{m}$, crosses zero at approximately $372 \mu\text{m}$, and rises to +1.751% ($\Delta K \approx +2,203 \text{ N/m}$) at $1004 \mu\text{m}$, corresponding to a total swing of 2.66 percentage points. The monolithic beam ($K_{0,M}$) shows a monotonically softening profile at the same temperature, with $\Delta K/K_0$ falling from -0.03% ($\Delta K \approx -61 \text{ N/m}$) to -1.22% ($\Delta K \approx -2,485 \text{ N/m}$) over the same amplitude range. The zero-crossing in the jointed data marks the displacement at which contact hardening exactly balances geometric softening; no equivalent crossover exists in the monolithic data at any temperature.

3.4 Temperature Dependence of the Backbone

Figure 6 shows the backbone curves of the jointed (Figure 6a) and monolithic (Figure 6b) specimens at four representative temperatures (-30°C, 20°C, 70°C, 120°C). To isolate the amplitude-dependent nonlinear

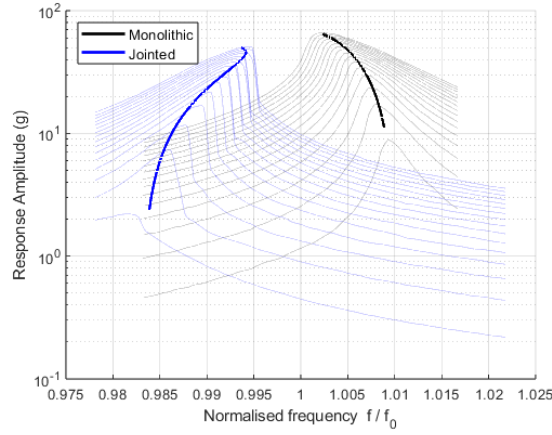


Figure 5: Backbone curves at 20°C for the jointed (blue) and monolithic (black) specimens on a normalised frequency axis (f/f_0), showing opposing nonlinear characters attributable to the bolted interface.

character from the baseline frequency shift driven by temperature-dependent material stiffness, the backbone curves are presented on a normalised frequency axis f/f_0 , where f_0 is the low-amplitude resonance frequency at each temperature.

Two observations are immediately apparent. First, the direction of backbone tilt reverses between -30°C (leftward, softening) and 20°C (rightward, hardening), demonstrating that the dominant nonlinear mechanism changes with temperature. Second, the magnitude of the tilt, and hence the strength of the nonlinearity, is non-monotonic with temperature, peaking near 20-30°C before reducing toward the thermal extremes. Neither behaviour is present in the monolithic specimen, which tilts leftward at all temperatures demonstrating consistently softening behaviour with relatively temperature-insensitive tilt magnitude, confirming that both effects originate at the bolted interface. The reversal in tilt direction between -30°C and 20°C for the jointed specimen indicates a thermally-gated transition in the dominant nonlinear mechanism.

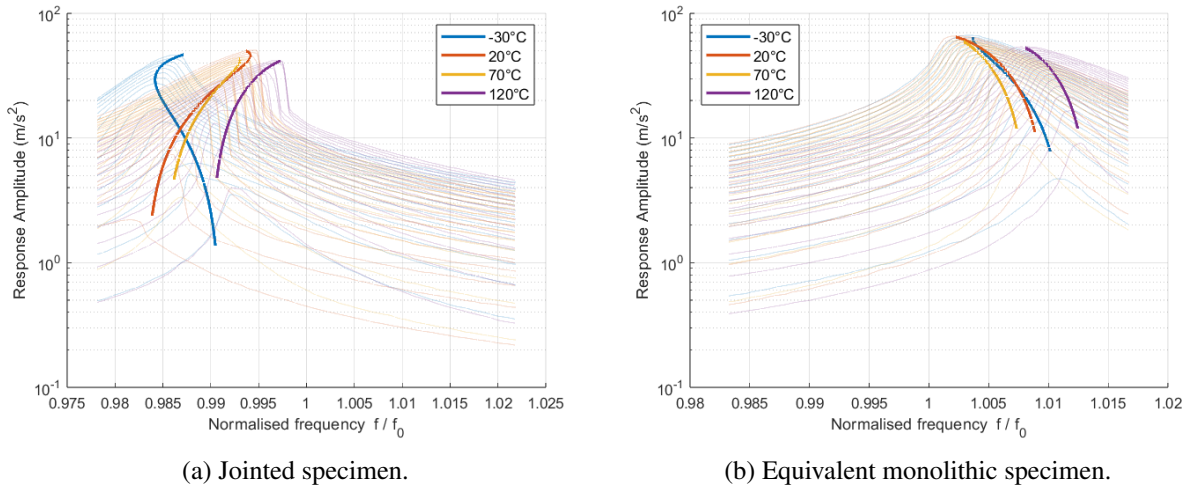


Figure 6: Backbone curves of the jointed (left) and monolithic (right) specimens at -30°C, 20°C, 70°C, and 120°C on a normalised frequency axis f/f_0 .

The backbone frequency shift (BFS), defined as:

$$\text{BFS} = \frac{f_{\text{res,max}} - f_{\text{res,min}}}{f_{\text{res,min}}} \times 100\% \quad (3)$$

quantifies the tilt magnitude across the full amplitude range and is summarised in Table 1 and Figure 7. For

the monolithic beam, BFS was negative at all 16 temperatures (range -0.331% to -0.728%; mean -0.534%, standard deviation $\pm 0.047\%$), confirming softening at every condition. For the jointed beam, BFS changed sign between -10°C (-0.352%) and 0°C (+0.176%), peaked at +1.334% near 20°C, and declined to +0.440% at 120°C (standard deviation $\pm 0.060\%$). This sign reversal is reproducible across independent experimental runs to three significant figures.

Table 1: Backbone frequency shift and amplitude-dependent stiffness deviation for both systems across the tested temperature range. Column 7 denotes the ΔK zero-crossing displacement of the jointed beam; dashes indicate no crossing within the tested amplitude range.

T [°C]	Monolithic		Jointed			Cross. [μm]
	BFS [%]	$\Delta K(0.022 \text{ g})$ [%]	BFS [%]	$\Delta K(0.384 \text{ g})$ [%]	$\Delta K(0.384 \text{ g})$ [%]	
-30	-0.728	-0.955	-0.616	+0.925	-0.315	309
-20	-0.465	-0.822	-0.176	+0.568	+0.215	248
-10	-0.530	-0.821	-0.352	+1.281	+0.570	201
0	-0.530	-0.690	+0.176	+0.570	+0.923	397
10	-0.663	-0.955	+0.887	-0.668	+1.103	446
20	-0.728	-1.217	+1.334	-1.196	+1.457	372
30	-0.664	-1.350	+1.333	-1.020	+1.635	360
40	-0.598	-1.350	+1.332	-0.844	+1.814	348
50	-0.597	-1.349	+1.065	-0.493	+1.637	312
60	-0.531	-1.219	+0.885	-0.315	+1.457	271
70	-0.465	-1.085	+0.620	-0.138	+1.103	223
80	-0.464	-0.822	+0.709	-0.492	+0.925	236
90	-0.398	-0.558	+0.797	-0.492	+1.101	220
100	-0.331	-0.293	+0.620	-0.138	+1.103	196
110	-0.397	-0.161	+0.529	+0.393	+1.457	165
120	-0.462	-0.028	+0.440	+0.925	+1.816	323

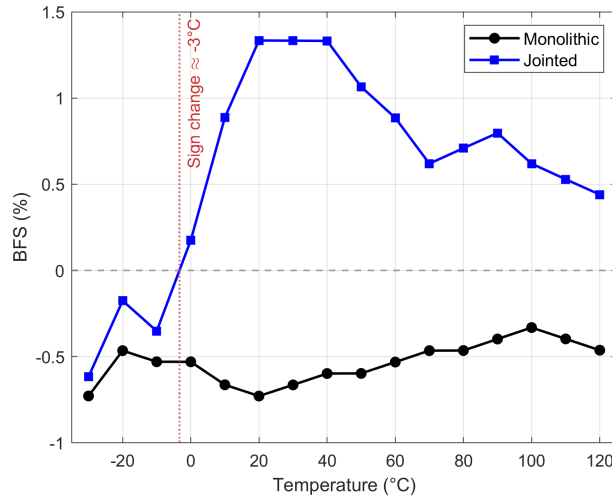


Figure 7: BFS versus temperature for the jointed (blue) and monolithic (black) specimens. The sign change in the jointed curve near 0°C marks the transition from softening to hardening-dominated nonlinear character.

The non-monotonic BFS profile of the jointed beam, rising from near zero at cold temperatures to a peak near 20-30°C before declining, reflects two competing thermally-driven mechanisms acting simultaneously on the contact interface, which are developed in full in Sections 3.2 and 3.5.

3.4.1 Monolithic Beam

For the monolithic beam, $R^2 > 0.83$ at all 16 temperatures (mean $R^2 = 0.95$), confirming that a single cubic softening term describes the backbone with high fidelity across the full thermal range. The fitted $\alpha_M(T)$ was negative at every temperature, ranging from $-3.0 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ at -20°C to $-6.0 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ at 120°C .

The physical origin is von Kármán membrane-bending coupling intrinsic to clamped-clamped beams: mid-plane stretching under transverse deflection generates a nonlinear restoring force proportional to the cube of displacement [19], producing a negative Duffing coefficient. Thermal loading modulates this geometric nonlinearity through the axial pre-stress imposed by the clamped boundary conditions. Below the thermal neutral point, the temperature at which thermally-induced axial stress is zero, the beam is under compression, which reduces the geometric stiffness contribution (analogous to the reduction in lateral stiffness of a column under compressive axial load below its critical load) and amplifies the softening coefficient. Above the neutral point, tensile axial stress partially counteracts geometric softening. The neutral point is identified from the data as the temperature where ΔK at the lowest tested amplitude changes sign: interpolation between $+0.238\%$ at 20°C and -0.028% at 30°C places it at approximately 29°C . This thermal modulation of a purely geometric nonlinearity is itself an instance of coupled thermal-mechanical behaviour, present even without a contact interface.

3.4.2 Jointed Beam

For the jointed beam, R^2 varied strongly with temperature, and this variation is itself mechanistically informative. At -30°C , $R^2 = 0.12$; at -20°C , $R^2 = 0.03$; the backbone at these temperatures cannot be described by a single cubic term, indicating that two competing mechanisms of comparable magnitude produce a response that is not well-approximated by a single power law. At these temperatures the geometric softening and contact hardening contributions are of comparable magnitude but opposite sign across the tested amplitude range, producing a backbone shape that is neither monotonically softening nor hardening. A higher-order (quintic or piecewise) stiffness model would be required to capture both contributions simultaneously, though fitting such a model reliably from the available data points is beyond the scope of the present study. At 0°C and above, R^2 ranged from 0.73 to 0.92, indicating that one mechanism becomes dominant and the Duffing approximation is adequate.

The fitted $\alpha_J(T)$ changed sign at approximately -22°C (interpolated between $-1.0 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ at -30°C and a near-zero positive value at -20°C), and peaked at $+3.0 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ near 50°C before declining to $+2.0 \cdot 10^{-4} \text{ Hz}^2/\mu\text{m}^2$ at 120°C . The non-monotonic temperature profile of α_J reflects two competing thermal influences on the contact hardening mechanism, which are discussed in Section 3.5.

3.5 Contact Mechanics and Thermally-Modulated Micro-slip

3.5.1 Micro-slip Hardening Mechanism

The amplitude-dependent stiffening of the jointed beam above the ΔK zero-crossing is attributed to progressive micro-slip at the bolted interface. This interpretation is grounded in the Mindlin contact model [20], which describes the tangential compliance of conforming elastic contacts under combined normal and oscillating tangential loads.

In the partial slip regime, an annular slip zone nucleates at the contact periphery (where the tangential traction first reaches the Coulomb limit) and grows inward as the tangential load amplitude increases. The energy dissipated per cycle scales as Q^3/P^2 [21], where Q is the tangential force amplitude and P is the contact normal preload. As the displacement amplitude grows, the slip annulus expands while a stuck central zone persists. The key consequence for global structural stiffness is that load redistribution from the slipping periphery to the stuck centre progressively increases the proportion of the interface carrying load through high-stiffness stuck contact, rather than through low-stiffness frictional sliding. This net effect on the global load path is an increase in effective structural stiffness with amplitude, the hardening character observed above the ΔK zero-crossing [20].

This mechanism is distinct from the geometric softening of the monolithic beam: the latter is governed by beam kinematics, whereas micro-slip hardening arises from the evolving contact state at the interface. Their superposition in the jointed beam produces the characteristic dip-then-rise ΔK profile, in which geometric softening dominates at small amplitudes (below the partial slip threshold) and contact hardening dominates at large amplitudes.

3.5.2 Thermal Modulation of Preload and Its Effect on Nonlinearity

The critical insight of this study is that temperature modulates the micro-slip hardening mechanism through the bolt preload. Since the bolt and beam are made from the same material (mild steel), the dominant preload variation arises not from differential thermal expansion but from the axial restraint imposed by the clamped boundary conditions: thermal expansion of the beam is resisted by the clamps, inducing a compressive axial force below the thermal neutral point and a tensile force above it. This axial force acts on the bolt-clamped interface and modifies the effective contact preload P . As an order-of-magnitude estimate, converting the applied joint torque of 1.5 Nm to a bolt preload via $P = \tau/(Kd)$ with nut factor $K = 0.20$ and nominal diameter $d = 5$ mm gives $P \approx 1.5$ kN. The axial force induced in the beam by a 10°C temperature change is $F = EA\alpha\Delta T = 200 \times 10^9 \times (9 \times 10^{-5}) \times 11.7 \times 10^{-6} \times 10 = 2.1$ kN, comparable in magnitude to the bolt preload itself, confirming that thermally-induced axial forces are sufficient to produce a measurable change in contact normal force across the tested temperature range.

The relationship between temperature and joint preload P is governed by the competing thermal deformations of the bolt and clamped members. For a joint assembled from a single material, the change in preload due to a uniform temperature change ΔT can be expressed as:

$$\Delta P = \frac{(\alpha_b - \alpha_c)\Delta T}{\frac{1}{k_b} + \frac{1}{k_c}} \quad (4)$$

where α_b and α_c are the coefficients of thermal expansion of the bolt and clamped members respectively, and k_b and k_c are their axial stiffnesses. Since both are steel, equation (4) predicts $\Delta P \approx 0$ for uniform heating, confirming that the dominant preload variation arises from the geometric asymmetry of the joint rather than material mismatch.

At low temperatures, thermal contraction reduces the compressive force at the joint interface, lowering P and hence lowering the partial slip threshold. Micro-slip initiates at smaller displacement amplitudes, but the reduced preload also limits the maximum tangential stiffness achievable in the stuck zone, suppressing the hardening contribution. At high temperatures, thermal expansion increases P , raising the slip threshold and expanding the amplitude range over which the stuck-zone stiffening operates, strengthening the hardening term.

This two-way coupling, temperature controlling preload and preload controlling the onset and magnitude of micro-slip hardening, is the fundamental thermal-mechanical interaction quantified in this study. It explains three experimental observations simultaneously:

1. **The BFS sign change near 0°C :** below this temperature, reduced preload suppresses hardening sufficiently that geometric softening dominates the net response; above it, increased preload allows hardening to dominate.
2. **The non-monotonic $\alpha_J(T)$ profile:** at moderate temperatures (0 - 60°C), increasing preload expands the partial slip amplitude range, strengthening α_J . Above 60°C , the slip threshold exceeds the tested amplitude range, so a progressively smaller fraction of the backbone is governed by micro-slip hardening and α_J plateaus.
3. **The ΔK zero-crossing shift:** from $\sim 446 \mu\text{m}$ at 10°C , decreasing to $\sim 165 \mu\text{m}$ at 110°C , consistent with elevated temperature increasing contact preload and thereby increasing the tangential stiffness of the stuck zone. Although higher preload raises the slip threshold, the accompanying increase in

stuck-zone stiffness amplifies the hardening contribution more strongly than it delays its onset, so the amplitude at which contact hardening first overcomes geometric softening decreases with temperature.

To test quantitative consistency with the Mindlin framework, the power law $\Delta K \propto A^n$ was fitted in log-space to the jointed ΔK values above the zero-crossing at each temperature. The fitted exponent n ranged from 1.35 to 3.69 with a valid zero-crossing (mean 2.04 ± 0.85). The Mindlin partial slip model predicts $n \approx 0.67$; gross slip raises the effective exponent toward $n = 2$. The observed range $1 \lesssim n \lesssim 2$ across most temperatures is intermediate between these limits, indicating that the interface operates in a mixed partial-to-gross slip regime across the tested displacement range (41-1018 μm). The systematic temperature-dependence of n , lower near 50-70°C and higher at the thermal extremes, is consistent with the contact normal force shifting the partial-to-gross slip boundary relative to the tested range as temperature varies. These exponents represent an effective structural scaling rather than a direct contact measurement, but their magnitude and systematic variation provide quantitative support for the micro-slip interpretation.

3.5.3 Thermally-Gated Nonlinear Transition

The reversal in net nonlinear character near 0°C is independently confirmed by two observables: the BFS sign change (between -10°C and 0°C) and the cubic modal stiffness coefficient zero-crossing ($\approx -22^\circ C$). These temperatures do not coincide exactly because each metric captures a different aspect of the response. The BFS integrates behaviour across the full amplitude range; the onset displacement is most sensitive to the low-amplitude threshold between stuck and partial-slip regimes; and the α_J zero-crossing weights the high-amplitude end of the backbone more heavily due to the A^2 dependence in Equation (2). The offset between $\approx -22^\circ C$ and $0^\circ C$ therefore reflects these amplitude-weighting differences rather than experimental inconsistency. Collectively, these observables identify a thermal transition in the range $-30^\circ C$ to $0^\circ C$ at which the dominant nonlinear mechanism shifts from geometric softening to contact micro-slip hardening.

3.6 Implications for Modelling and Design

These results demonstrate that the nonlinear dynamic behaviour of a bolted joint under combined thermal and mechanical loading cannot be characterised at a single operating condition. The Duffing coefficient of the jointed system changes sign across the tested thermal range, transitioning from softening to hardening near $-22^\circ C$. A model calibrated at $20^\circ C$ will predict hardening with $\alpha_J = +1.71 \cdot 10^{-4} \text{ Hz}^2/\mu m^2$; the same structure at $-30^\circ C$ exhibits net softening with $\alpha_J = -1.0 \cdot 10^{-4} \text{ Hz}^2/\mu m^2$. For spacecraft structures subjected to orbital thermal cycling, where the operational envelope of $-40^\circ C$ to $+80^\circ C$ spans the identified $0^\circ C$ transition, the nonlinear character of jointed components may reverse on each orbit. Single-temperature characterisation is demonstrably insufficient to predict behaviour across such an envelope, and any predictive model must incorporate the coupled dependence of interface stiffness on both displacement amplitude and temperature simultaneously.

4 Conclusion

This study has experimentally characterised the coupled thermal-mechanical nonlinear dynamic behaviour of a clamped-clamped bolted lap-joint beam across 16 temperatures (from -30 to $+120^\circ C$) and 16 excitation amplitudes (0.022-0.384 g RMS; displacement range 41-1018 μm), using an equivalent monolithic beam as a controlled baseline to isolate joint-induced behaviour.

Impact hammer testing established that the bolted lap joint reduces the fundamental modal stiffness by 38.2% (from $k_M = 203.7 \text{ kN/m}$ to $k_J = 125.8 \text{ kN/m}$), which was confirmed self-consistently by the ratio $k_J/k_M = (m_J/m_M)(f_J/f_M)^2 = 0.618$. This linear, amplitude-independent effect is fully accounted for by joint interface compliance and the 1.9% difference in physical mass between the configurations.

The monolithic beam exhibited a softening Duffing nonlinearity at all temperatures (BFS -0.331% to -0.728% ; $R^2 > 0.83$). Thermal loading modulates this purely geometric nonlinearity through axial pre-stress

imposed by the clamped boundary conditions, with a neutral point at approximately 29°C separating compressive and tensile regimes. This constitutes coupled thermal-mechanical behaviour even without a contact interface.

The jointed beam exhibited a qualitatively distinct response governed by the superposition of geometric softening and contact micro-slip hardening. Below approximately 372 μm displacement at 20°C, geometric softening dominates ($\Delta K < 0$); above this threshold, progressive micro-slip at the bolted interface produces a net hardening effect ($\Delta K > 0$). The net nonlinear character reverses sign at 0°C: BFS is negative at $T < 0^\circ\text{C}$ and positive at $T \geq 0^\circ\text{C}$, peaking at +1.334% near 20–30°C. This thermally-gated transition is corroborated independently by the cubic modal stiffness coefficient zero-crossing interpolated at $\approx -22^\circ\text{C}$. Both observables are consistent with 0°C marking the temperature at which thermally-induced axial stress transitions from compressive to tensile, simultaneously relieving geometric softening and increasing joint contact preload.

These findings directly demonstrate that the nonlinear character of a bolted joint is fundamentally coupled to temperature and cannot be characterised at a single temperature or amplitude level. A model calibrated at ambient temperature may predict hardening behaviour, but the same structure at -30°C exhibits net softening. For structures subject to wide thermal excursions, such as spacecraft experiencing orbital thermal cycling, single-condition characterisation is insufficient to predict behaviour across the operational envelope.

The primary limitation is that the results are drawn from a single specimen of each configuration, so joint-to-joint variability arising from differences in bolt torque, surface finish, or assembly history has not been assessed. Future work should replicate the experiment across multiple assemblies to establish the statistical robustness of the identified transition temperatures and crossover thresholds, incorporate direct measurement of interface contact force to replace the current inferential interpretation of the thermal pre-stress effect, and extend the analysis to multi-bolt configurations and higher-order modes to generalise the findings toward industrially representative assemblies.

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